SSC CGL 2024 Tier-I question paper with answers

19 September 2024 · Shift 3

74 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: +2 for a right answer, –0.50 for a wrong one

The other 26 of the paper's 100 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/Writeup_Final_Anwerkey_CGLE_2024_T1_191224.pdf

Compare your score: the official 2024 cut-off (Tier-I, all posts except JSO and SI Grade-II), out of 200

UR 153.19 **OBC** 146.26 **EWS** 142.02 **SC** 126.46 **ST** 111.89 **ESM** 69.93 **OH** 113.10 **HH** 64.79 **VH** 102.97 **Other PwD** 45.74

Cut-offs apply to marks normalised across all the exam's shifts, so a raw score on one shift is a rough guide. Source: the official result write-up, https://ssc.gov.in/api/attachment/uploads/masterData/NoticeBoards/Result%20Write-up%20CGLE%202024%20Tier%20I_51224.pdf

General Intelligence and Reasoning 17 questions

Q.2

In a certain code language, ‘CYCLE’ is coded as ‘242241522’ and ‘FINAL’ is coded as ‘2118132615’. How will ‘MUSIC’ be coded in that language?

(A) 14681824
(B) 1421191824
(C) 146191824
(D) 146192184

Q.3

Four letter-clusters have been given, out of which three are alike in some manner and one is different. Select the letter-cluster that is different. Note: The odd one out is not based on the number of consonants/vowels or their position in the letter cluster...

(A) TVX
(B) HJL
(C) MOP
(D) NPR

Q.4

Select the triad in which the numbers are related to each other in the same way as the numbers in the following triads. 4-5-41 2-3-13 (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 7-8-110
(B) 10-11-211
(C) 12-13-213
(D) 5-6-61

Q.5

In a certain code language, ‘PANTHER’ is written as ‘0146283’ and ‘ANOTHER’ is written as ‘2864109’. How will ‘O’ be written in that language?

(A) 3
(B) 8
(C) 0
(D) 9

Q.7

'MPQO' is related to 'KNSQ' in a certain way based on the English alphabetical order. In the same way, 'DGHF' is related to 'BEJH'. To which of the following is 'EINW' related, following the same logic?

(A) CHQY
(B) BHQX
(C) BGPX
(D) CGPY

Q.8

Select the number from the given options to complete the series. 58, 70, 83, 97, ___

(A) 112
(B) 113
(C) 114
(D) 111

Q.9

In a certain code language, X + Y means 'X is the sister of Y', X – Y means 'X is the wife of Y', X × Y means 'X is the mother of Y', X ÷ Y means 'X is the father of Y'. Based on the above, how is H related to L if 'H × I – J ÷ K + L'?

(A) Father's mother
(B) Mother
(C) Mother's mother
(D) Father's sister

Q.10

Select the word-pair that best represents a similar relationship to the one expressed in the pair of words given below. (The words must be considered as meaningful English words and must not be related to each other based on the number of letters/number of consonants/vowels in the word.) Larva : Insect

(A) Dog : Animal
(B) Fawn : Deer
(C) Fox : Vixen
(D) Bat : Chick

Q.12

What will come in place of the question mark (?) in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 69 + 2485 × 5 – 912 ÷ 2 = ?

(A) 1396
(B) 1936
(C) 1639
(D) 1963

Q.14

Select the triad in which the numbers are related in the same way as are the numbers of the given triads. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g., 13 – Operations on 13 such as adding/deleting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (11, 1, 10) (13, 7, 90)

(A) (1, 2, 15)
(B) (4, 5, 19)
(C) (2, 6, 28)
(D) (3, 6, 23)

Q.15

Three of the following four options are alike in a certain way and thus form a group. Which is the one that does NOT belong to that group? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 88 – 23 – 245
(B) 56 – 31 – 205
(C) 22 – 13 – 85
(D) 18 – 12 – 72

Q.16

Select the option in which the numbers share the same relationship as that shared by the given pair of numbers. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/deleting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) 11 : 62 13 : 64

(A) 12 : 92
(B) 9 : 59
(C) 14 : 37
(D) 16 : 67

Q.17

What will come in the place of '?' in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 49 × 7 – 6 ÷ 8 +13 =

(A) 42
(B) 37
(C) 43
(D) 32

Q.19

In this question, three statements are given, followed by two conclusions numbered I and II. Assuming the statements to be true, even if they seem to be at variance with commonly known facts, decide which of the conclusion(s) logically follows/follow from the statements. Statements: Some shirts are pants. All pants are jackets. All jackets are trousers. Conclusions: I. Some trousers are shirts. II. All jackets are shirts.

(A) Only conclusion I follows.
(B) Only conclusion II follows.
(C) Both conclusions I and II follow.
(D) Neither conclusion I nor II follows.

Q.21

Which of the following terms will replace the question mark (?) in the given series? HDLB, LBPZ, PZTX, ? , XVBT

(A) TWXC
(B) TXXV
(C) TWWV
(D) TVWC

Q.24

The position of how many letters will remain unchanged if each of the letters in the word 'DRAFTING' is arranged in the English alphabetical order?

(A) None
(B) One
(C) Three
(D) Two

Q.25

Select the option that represents the letters that when sequentially placed from left to right in the blanks below will complete the letter series. I _ _ I _ I _ _ I _ J _

(A) J I J I J I I
(B) J I I I J J I
(C) I J I I J J I
(D) I I J J J I J

General Awareness 21 questions

Q.1

Which of the following is the largest inhabited riverine island in the world?

(A) Lakshadweep
(B) Peacock
(C) Majuli
(D) Bhavani

Q.2

In the Right-Hand Thumb Rule, the thumb is directed towards the direction of:

(A) current
(B) magnetic field
(C) electric field
(D) motion of the conductor

Q.4

Which is the most important protein component in milk, both quantitatively and nutritionally, that accounts for about 80% of the total protein in bovine milk?

(A) Pepsin
(B) Casein
(C) Actin
(D) Albumin

Q.5

Which web portal was launched by the Department of Telecommunications for sharing information on mobile towers and EMF emission compliance?

(A) E-Amrit
(B) Tarang Sanchar
(C) Namaste
(D) PM-Daksh

Q.7

In 1539, the Battle of Chausa was fought between Humayun and _____.

(A) Ibrahim Lodhi
(B) Sikandar Suri
(C) Sher Shah Suri
(D) Rana Sanga

Q.9

Sardhana Christian Fair is held in which of the following states?

(A) Haryana
(B) Uttar Pradesh
(C) Madhya Pradesh
(D) West Bengal

Q.10

Ali Akbar Khan was associated with which of the following gharanas?

(A) Kirana gharana
(B) Gwalior gharana
(C) Agra gharana
(D) Maihar gharana

Q.12

Which Article of the Indian Constitution mentions that ‘The State shall endeavour to promote cottage industries on an individual or co-operative basis in rural areas’?

(A) Article 42
(B) Article 44
(C) Article 49
(D) Article 43

Q.13

Which of the following is the most controversial Article in the Indian Constitution, which provides for President’s rule in any state?

(A) Article 356
(B) Article 358
(C) Article 352
(D) Article 360

Q.14

A dense mass of water drops on smoke or dust particles in the lower atmosphere layers is referred to as:

(A) smog
(B) frost
(C) blizzard
(D) mist

Q.15
Bhakti Pradeep Kulkarni is associated with _____.

(A) cricket
(B) kabaddi
(C) chess
(D) badminton

Q.16
Who among the following was the youngest recipient of the Sangeetha Kalanidhi awarded by the Music Academy in 1947?

(A) Bhimsen Joshi
(B) Ariyakudi Ramanuja Iyengar
(C) Mallikarjun Bheemarayappa Mansur
(D) Semmangudi Radhakrishna Srinivasa Iyer

Q.17
In which year was the battle of Buxar fought?

(A) 1777
(B) 1793
(C) 1764
(D) 1757

Q.18
Which of the following State governments has launched a scheme 'Ankur' in May 2021 to promote planting of trees in public, wherein people will be awarded for planting and taking care of the saplings?

(A) West Bengal
(B) Bihar
(C) Uttarakhand
(D) Madhya Pradesh

Q.19
Which of the following Acts passed under Warren Hastings was also called 'half-loaf system'?

(A) Pitt's India Act 1784
(B) Government of India Act 1858
(C) Regulating Act 1773
(D) Charter Act 1813

Q.20
In Newlands' Octaves, the properties of lithium and _____ were found to be the same.

(A) sodium
(B) beryllium
(C) aluminium
(D) magnesium

Q.21
India hosted the Asian Games in which of the following years?

(A) 1954 and 1982
(B) 1951 and 1986
(C) 1951 and 1982
(D) 1954 and 1986

Q.22
Which of the following movements created regional disparities between large and small farmers in India?

(A) Yellow revolution
(B) Green revolution
(C) Pink revolution
(D) Blue revolution

Q.23
What was the main objective of the Second Green Revolution in India?

(A) Ensuring food security in the western states
(B) Focusing on holistic development in agriculture
(C) Increasing the use of chemical fertilizers and pesticides
(D) Promoting conventional agricultural practices

Q.24

Which of the following Articles of the Indian Constitution CANNOT be suspended even during an emergency declared on the grounds of war or external aggression?

(A) Articles 15 and 17
(B) Articles 20 and 21
(C) Articles 14 and 16
(D) Articles 30 and 32

Q.25

Elelakkaradi dance of Kerala is performed by which of the following tribes?

(A) Mavilan tribe
(B) Kurumba tribe
(C) Irular tribe
(D) Adiyan tribe

Quantitative Aptitude 11 questions

Q.1

Two inlet pipes, P1 and P2, can fill a cistern in 20 hours and 30 hours, respectively. They were opened at the same time, but pipe P1 had to be closed 5 hours before the cistern was full. How many hours in total did it take for the two pipes to fill the cistern?

(A) 15 hours
(B) 9 hours
(C) 12 hours
(D) 10 hours

Q.3

The ratio of boys to girls initial in a college is 5 : 6. If 40 boys leave the college and 50 girls join the college, the ratio will become 8 : 11. Find the number of girls (initial) in the college.

(A) 740
(B) 640
(C) 720
(D) 860

Q.4

From a point A, a tangent line is drawn to the circle of radius 7 units. From the same point, a secant is drawn to the circle which cuts the circle at B and C and the point B is near A than C. What is the length of BC in units if the length of the tangent to the circle from point A is 21 units and the length of AB is 14 units?

(A) 18.5
(B) 16.5
(C) 15.5
(D) 17.5

Q.6

Pranav borrows a sum of ₹598397 at the rate of 20% per annum simple interest. At the end of the first year, he repays ₹24284 towards return of principal amount borrowed. If Pranav clears all pending dues at the end of the second year, including interest payment that accrued during the first year, how much does he pay (in ₹) at the end of the second year?

(A) 808615
(B) 812514
(C) 817567
(D) 818513

Q.7

A class of 30 students appeared in a test. The average score of 12 students is 62, and that of the rest is 68. What is the average score of the class?

(A) 66.6
(B) 63.6
(C) 65.6
(D) 64.6

Q.8

If the area of a sector of a circle is 462 cm^2 and the central angle is 120°, then its radius is ____.

(A) 18 cm
(B) 17 cm
(C) 14 cm
(D) 21 cm

Q.14

Which of the following numbers is NOT divisible by 8?

(A) 5544
(B) 7344
(C) 7497
(D) 4608

Q.20

A cooker is marked at ₹12,500. The dealer allows successive discounts of 6%, 6% and 4% on it. What is the net selling price of the cooker?

(A) ₹12,320.60
(B) ₹10,306
(C) ₹10,603.20
(D) ₹12,230

Q.21

A thief is spotted by a policeman from a distance of 195 metres. When the policeman starts the chase, the thief also starts running. If the speed of the thief is 22 km/h and that of the policeman is 27 km/h, then how far would the thief have run (in m) before he is overtaken?

(A) 858
(B) 958
(C) 951
(D) 825

Q.23

In a kite ABCD, longer diagonal BD is drawn. If ∠ABD = 54° and ∠ADB = 50°, then what is the measure (in degrees) of the ∠DCB?

(A) 108
(B) 100
(C) 54
(D) 76

Q.24

Three candidates X, Y and Z participated in an election. X got 30% more votes than Y, whereas Z got 25% more votes than Y. X also overtook Z by 5000 votes. If 71% of the voters voted and no invalid votes were cast, then what was the total number of voters in the voting list?

(A) 5,00,000
(B) 4,50,000
(C) 3,55,000
(D) 4,05,000

English Comprehension 25 questions

Q.1

Select the most appropriate ANTONYM of the given word. Appreciate

(A) Acknowledge
(B) Depreciate
(C) Depict
(D) Esteem

Q.2

Identify the INCORRECTLY spelt word in the following sentence and select its correct spelling from the given options. The occurence of this phenomenon is quite rare.

(A) occurrence
(B) occurance
(C) occerence
(D) occurrance

Q.3

Select the option that expresses the given sentence in passive voice. The school will hire new teachers.

(A) New teachers will join the school.
(B) New teachers are hired by the school.
(C) New teachers will be hired by the school.
(D) The school will be hiring new teachers.

Q.4

Select the most appropriate synonym of the given word. Magnificent

(A) Sacred
(B) Scared
(C) Splendid
(D) Slow

Q.5

Select the most appropriate option to fill in the blank. The US _____ building was built in the year 1800.

(A) captain
(B) capitol
(C) capital
(D) cannibal

Q.6

Parts of the following sentence have been given as options. Select the option that contains an error. In some countries, you are not able to drink until you are 21.

(A) until you are 21.
(B) able to drink
(C) you are not
(D) In some countries,

Q.7

Select the term which means the same as the given group of words. A continuing unpleasant situation, created when one problem causes another problem that then makes the first problem worse

(A) Round cycle
(B) Repetitive cycle
(C) Vicious cycle
(D) Enclosed cycle

Q.8

Select the option that corrects the error in the given sentence. She sings good.

(A) better
(B) goodly
(C) best
(D) well

Q.9

Parts of a sentence are given below in jumbled order. Select the option that arranges the parts in the correct sequence to form a meaningful sentence. (P) society benefits from diversity (Q) enriching cultures and (O) by fostering understanding and empathy (R) perspectives from around the world

(A) QPRO
(B) POQR
(C) ROQP
(D) RPOQ

Q.10

Select the option that can be used as a one-word substitute for the given group of words. Lack of interest, enthusiasm or concern

(A) Apathy
(B) Antipathy
(C) Sympathy
(D) Empathy

Q.11

Select the option that can be used as a one-word substitute for the given group of words. The study of ancient things such as art, graves, ruins etc.

(A) Astrology
(B) Meteorology
(C) Archaeology
(D) Geology

Q.12

Select the most appropriate option that can substitute the underlined segment in the given sentence. No sooner did Mr. Piyahri Mishra finish his chemistry lecture “that the students began to leave”.

(A) than the students began to leave.
(B) then the students began to leave.
(C) so the students began to leave.
(D) and the students began to leave.

Q.13

Select the option that can be used as a one-word substitute for the given group of words. An utter state of confusion

(A) Chaos
(B) Pathos
(C) Innuendos
(D) Engross

Q.14

Select the INCORRECTLY spelt word.

(A) Cemetery
(B) Occurrence
(C) Beggining
(D) Interrupt

Q.15

Select the most appropriate synonym of the given word. Miraculous

(A) Expected
(B) Average
(C) Incredible
(D) Draining

Q.16

Select the most appropriate option to substitute the underlined segment in the given sentence. After reflecting on the matter regarding his boss's selfish attitude, Raju decided to "force himself to perform an unpleasant action in a difficult situation in order to resolve the matter."

(A) dodge the bullet
(B) ignore the bullet
(C) bite the bullet
(D) shoot the bullet

Q.17

Select the most appropriate ANTONYM of the given word. Absolve

(A) Appeal
(B) Accuse
(C) Excuse
(D) Commend

Q.18

Select the option that expresses the given sentence in passive voice. Meenakshi could have sent a reply.

(A) A reply could have been send by Meenakshi.
(B) A reply could have be sent by Meenakshi.
(C) A reply could have been sent by Meenakshi.
(D) A reply could have sent by Meenakshi.

Q.19

Select the most appropriate synonym of the given word. Beseech

(A) Impart
(B) Implore
(C) Bequeath
(D) Pamper

Q.20

Select the option that can be used as a one-word substitute for the given group of words. A person who has supreme power or authority

(A) Foreign
(B) Secular
(C) Diversity
(D) Sovereign

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Augmented reality and virtual reality are two types of reality technologies that either augment or replace a real-world environment with a (1)_____ one. In augmented reality, a virtual environment coexists (2)_____ the actual world, to be instructive and offer more data about the real world that a user can access without having to search. When a handset is pointed at a piece of malfunctioning (3)_____, for example, industrial AR apps might provide rapid troubleshooting information. Virtual reality is a (4)_____ environmental simulation that completely replaces the user's surroundings with a totally virtual world. For example, VR may provide a user with a box containing a cartoon (5)_____ of Mike Tyson in a virtual boxing ring.

Select the most appropriate option to fill in blank number 1.

(A) revealed
(B) simulated
(C) catered
(D) suffocated

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Augmented reality and virtual reality are two types of reality technologies that either augment or replace a real-world environment with a (1)_____ one. In augmented reality, a virtual environment coexists (2)_____ the actual world, to be instructive and offer more data about the real world that a user can access without having to search. When a handset is pointed at a piece of malfunctioning (3)_____, for example, industrial AR apps might provide rapid troubleshooting information. Virtual reality is a (4)_____ environmental simulation that completely replaces the user's surroundings with a totally virtual world. For example, VR may provide a user with a box containing a cartoon (5)_____ of Mike Tyson in a virtual boxing ring.

Select the most appropriate option to fill in blank number 2.

(A) beyond
(B) with
(C) along
(D) behind

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Augmented reality and virtual reality are two types of reality technologies that either augment or replace a real-world environment with a (1)_____ one. In augmented reality, a virtual environment coexists (2)_____ the actual world, to be instructive and offer more data about the real world that a user can access without having to search. When a handset is pointed at a piece of malfunctioning (3)_____, for example, industrial AR apps might provide rapid troubleshooting information. Virtual reality is a (4)_____ environmental simulation that completely replaces the user's surroundings with a totally virtual world. For example, VR may provide a user with a box containing a cartoon (5)_____ of Mike Tyson in a virtual boxing ring.

Select the most appropriate option to fill in blank number 3.

(A) equipment
(B) baggage
(C) outfit
(D) arsenal

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Augmented reality and virtual reality are two types of reality technologies that either augment or replace a real-world environment with a (1)_____ one. In augmented reality, a virtual environment coexists (2)_____ the actual world, to be instructive and offer more data about the real world that a user can access without having to search. When a handset is pointed at a piece of malfunctioning (3)_____, for example, industrial AR apps might provide rapid troubleshooting information. Virtual reality is a (4)_____ environmental simulation that completely replaces the user's surroundings with a totally virtual world. For example, VR may provide a user with a box containing a cartoon (5)_____ of Mike Tyson in a virtual boxing ring.

Select the most appropriate option to fill in blank number 4.

(A) comprehensive
(B) transgressive
(C) conclusive
(D) reprehensive

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Augmented reality and virtual reality are two types of reality technologies that either augment or replace a real-world environment with a (1)_____ one. In augmented reality, a virtual environment coexists (2)_____ the actual world, to be instructive and offer more data about the real world that a user can access without having to search. When a handset is pointed at a piece of malfunctioning (3)_____, for example, industrial AR apps might provide rapid troubleshooting information. Virtual reality is a (4)_____ environmental simulation that completely replaces the user's surroundings with a totally virtual world. For example, VR may provide a user with a box containing a cartoon (5)_____ of Mike Tyson in a virtual boxing ring.

Select the most appropriate option to fill in blank number 5.

(A) rendition
(B) glossary
(C) network
(D) resistance

Answer key

General Intelligence and Reasoning

Q.2 **(A)**	Q.3 **(C)**	Q.4 **(D)**	Q.5 **(D)**	Q.7 **(D)**	Q.8 **(A)**
Q.9 **(C)**	Q.10 **(B)**	Q.12 **(A)**	Q.14 **(B)**	Q.15 **(C)**	Q.16 **(D)**
Q.17 **(A)**	Q.19 **(A)**	Q.21 **(B)**	Q.24 **(A)**	Q.25 **(A)**	

General Awareness

Q.1 **(C)**	Q.2 **(A)**	Q.4 **(B)**	Q.5 **(B)**	Q.7 **(C)**	Q.9 **(B)**
Q.10 **(D)**	Q.12 **(D)**	Q.13 **(A)**	Q.14 **(A)**	Q.15 **(C)**	Q.16 **(D)**
Q.17 **(C)**	Q.18 **(D)**	Q.19 **(A)**	Q.20 **(A)**	Q.21 **(C)**	Q.22 **(B)**
Q.23 **(B)**	Q.24 **(B)**	Q.25 **(C)**			

Quantitative Aptitude

Q.1 **(A)**	Q.3 **(C)**	Q.4 **(D)**	Q.6 **(A)**	Q.7 **(C)**	Q.8 **(D)**
Q.14 **(C)**	Q.20 **(C)**	Q.21 **(A)**	Q.23 **(D)**	Q.24 **(A)**	

English Comprehension

Q.1 **(B)**	Q.2 **(A)**	Q.3 **(C)**	Q.4 **(C)**	Q.5 **(B)**	Q.6 **(B)**
Q.7 **(C)**	Q.8 **(D)**	Q.9 **(B)**	Q.10 **(A)**	Q.11 **(C)**	Q.12 **(A)**
Q.13 **(A)**	Q.14 **(C)**	Q.15 **(C)**	Q.16 **(C)**	Q.17 **(B)**	Q.18 **(C)**
Q.19 **(B)**	Q.20 **(D)**	Q.21 **(B)**	Q.22 **(B)**	Q.23 **(A)**	Q.24 **(A)**
Q.25 **(A)**					

Explanations

General Intelligence and Reasoning

Q.2 · Answer (A) 14681824

Each letter is coded by its place counted from Z (Z = 1, A = 26): C = 24, Y = 2, C = 24, L = 15, E = 22. So M = 14, U = 6, S = 8, I = 18, C = 24: 14681824.

Q.3 · Answer (C) MOP

TVX, HJL and NPR move +2, then +2. MOP moves +2, then +1, so it is the odd one.

Q.4 · Answer (D) 5-6-61

The third number is the sum of the squares of the first two: $4^2 + 5^2 = 41$ and $2^2 + 3^2 = 13$. Likewise $5^2 + 6^2 = 25 + 36 = 61$.

Q.5 · Answer (D) 9

A, N, T, H, E and R are in both words, as are 0, 1, 2, 4, 6 and 8 in both codes. The letter left in ANOTHER is O and the digit left in its code is 9, so O = 9.

Q.7 · Answer (D) CGPY

The first two letters move 2 back and the last two 2 forward: M → K, P → N, Q → S, O → Q. So E → C, I → G, N → P, W → Y: CGPY.

Q.8 · Answer (A) 112

The differences are 12, 13, 14, growing by 1, so the next is 15: 97 + 15 = 112.

Q.9 · Answer (C) Mother's mother

H is the mother of I, I is the wife of J, J is the father of K and K is the sister of L. So I is L's mother and H is I's mother: H is L's mother's mother.

Q.10 · Answer (B) Fawn : Deer

A larva is the young of an insect, as a fawn is the young of a deer. (A vixen is a female fox.)

Q.12 · Answer (A) 1396

After interchanging the signs: 69 – 2485 ÷ 5 + 912 × 2 = 69 – 497 + 1824 = 1396.

Q.14 · Answer (B) (4, 5, 19)

The third number is the product of the first two minus 1: 11 × 1 – 1 = 10 and 13 × 7 – 1 = 90. Likewise 4 × 5 – 1 = 19.

Q.15 · Answer (C) 22 – 13 – 85

In the others the third number is twice the first plus three times the second: 2 × 88 + 3 × 23 = 245, 2 × 56 + 3 × 31 = 205 and 2 × 18 + 3 × 12 = 72. For 22 – 13 it would be 44 + 39 = 83, not 85.

Q.16 · Answer (D) 16 : 67

The second number is the first plus 51: 11 + 51 = 62 and 13 + 51 = 64. Likewise 16 + 51 = 67.

Q.17 · Answer (A) 42

After interchanging the signs: 49 ÷ 7 + 6 × 8 – 13 = 7 + 48 – 13 = 42.

Q.19 · Answer (A) Only conclusion I follows.

Some shirts are pants, and all pants are jackets and so trousers, so some shirts are trousers: some trousers are shirts (I). Nothing says every jacket is a shirt, so II does not follow.

Q.21 · Answer (B) TXXV

The first and third letters move 4 forward (H, L, P, T, X and L, P, T, X, B) and the second and fourth 2 back (D, B, Z, X, V and B, Z, X, V, T): TXXV.

Q.24 · Answer (A) None

DRAFTING in alphabetical order is A D F G I N R T. Set against D R A F T I N G, no letter is in the same place.

Q.25 · Answer (A) J I J I J I I
With J, I, J, I, J, I, I in the blanks the block IJI repeats: IJI IJI IJI IJI.

General Awareness

Q.1 · Answer (C) Majuli
Majuli, in the Brahmaputra in Assam, is the world's largest inhabited river island.

Q.2 · Answer (A) current
Holding a straight current-carrying conductor in the right hand with the thumb along the current, the curled fingers give the direction of the magnetic field lines.

Q.4 · Answer (B) Casein
Casein makes up about 80% of the protein in cow's milk; most of the rest is whey protein.

Q.5 · Answer (B) Tarang Sanchar
The Department of Telecommunications' Tarang Sanchar portal shows the mobile towers in any locality and whether they comply with the EMF emission norms.

Q.7 · Answer (C) Sher Shah Suri
Sher Shah Suri defeated Humayun at Chausa in 1539, and again at Kannauj in 1540, driving him out of India.

Q.9 · Answer (B) Uttar Pradesh
Sardhana, near Meerut in Uttar Pradesh, is home to the Basilica of Our Lady of Graces built by Begum Samru, the centre of its annual Christian fair.

Q.10 · Answer (D) Maihar gharana
The sarod maestro Ali Akbar Khan was the son and disciple of Allauddin Khan, who shaped the Maihar gharana.

Q.12 · Answer (D) Article 43
Article 43, a Directive Principle, asks the State to secure a living wage for workers and to promote cottage industries in rural areas.

Q.13 · Answer (A) Article 356
Article 356 lets the President take over a state's government when its constitutional machinery fails; its frequent use has made it controversial. Article 352 covers a national emergency and Article 360 a financial one.

Q.14 · Answer (A) smog
SSC's key gives smog: fog mixed with smoke, its water droplets condensed on smoke and dust particles in the lower atmosphere. The description fits fog in general, which is not an option; 'dense' and 'smoke' point to smog rather than mist, which is lighter. Frost is ice deposited on surfaces, and a blizzard is a snowstorm.

Q.15 · Answer (C) chess
Bhakti Kulkarni of Goa is a chess Woman Grandmaster.

Q.16 · Answer (D) Semmangudi Radhakrishna Srinivasa Iyer
Semmangudi Srinivasa Iyer received the Music Academy's Sangeetha Kalanidhi in 1947, at 39 its youngest recipient. Bhimsen Joshi and Mallikarjun Mansur were Hindustani vocalists.

Q.17 · Answer (C) 1764
At Buxar in 1764 the East India Company defeated the combined forces of Mir Qasim, Shuja-ud-Daula and the Mughal emperor Shah Alam II.

Q.18 · Answer (D) Madhya Pradesh
Madhya Pradesh launched 'Ankur' in May 2021: people who plant a sapling and look after it can win the Pranvayu Award.

Q.19 · Answer (A) Pitt's India Act 1784
SSC's key gives Pitt's India Act of 1784, passed while Warren Hastings was Governor-General. It set up dual control: the Company's Court of Directors kept its commercial affairs, while a Board of Control appointed by the Crown supervised the government of its territories.

Q.20 · Answer (A) sodium
In Newlands' Octaves every eighth element resembles the first, as sodium, the eighth from lithium, does.

Q.21 · Answer (C) 1951 and 1982
New Delhi hosted the first Asian Games in 1951 and the ninth in 1982.

Q.22 · Answer (B) Green revolution
The Green Revolution's gains went mostly to large farmers, who could afford irrigation, high-yielding seeds and fertilisers, and to a few regions, widening the gaps.

Q.23 · Answer (B) Focusing on holistic development in agriculture
The Second Green Revolution looks beyond the first one's push for wheat and rice yields in a few north-western states, to holistic and sustainable development of agriculture as a whole.

Q.24 · Answer (B) Articles 20 and 21
Since the 44th Amendment (1978), Articles 20 and 21 cannot be suspended during a national emergency.

Q.25 · Answer (C) Irular tribe
Elelakkaradi is a group dance of the Irular (Irula) tribe of Kerala, in which men, women and children take part.

Quantitative Aptitude

Q.1 · Answer (A) 15 hours
If it took T hours, P1 ran for T − 5 of them: $(T - 5)/20 + T/30 = 1$, so $3T - 15 + 2T = 60$ and $T = 15$.

Q.3 · Answer (C) 720
With 5x boys and 6x girls: $(5x - 40)/(6x + 50) = 8/11$, so $55x - 440 = 48x + 400$, $x = 120$ and there were $6 \times 120 = 720$ girls.

Q.4 · Answer (D) 17.5
Tangent² = AB × AC, so $21^2 = 14 \times AC$ and AC = 31.5. BC = AC − AB = 31.5 − 14 = 17.5. (The radius is not needed.)

Q.6 · Answer (A) 808615
First-year interest: 20% of ₹598397 = ₹119679.40. After he repays ₹24284, ₹574113 is owed, with second-year interest of ₹114822.60. He pays 574113 + 119679.40 + 114822.60 = ₹808615.

Q.7 · Answer (C) 65.6
$(12 \times 62 + 18 \times 68)/30 = (744 + 1{,}224)/30 = 1{,}968/30 = 65.6$.

Q.8 · Answer (D) 21 cm
$(120/360) \times \pi r^2 = 462$, so $\pi r^2 = 1386$ and $r^2 = 1386 \times 7/22 = 441$: r = 21 cm.

Q.14 · Answer (C) 7497
7497 is odd, so it cannot be divisible by 8. The others are: 5544 = 8 × 693, 7344 = 8 × 918 and 4608 = 8 × 576.

Q.20 · Answer (C) ₹10,603.20
₹12,500 × 0.94 × 0.94 × 0.96 = ₹10,603.20.

Q.21 · Answer (A) 858
The policeman gains 27 − 22 = 5 km/h, so he closes 195 m in 0.195/5 = 0.039 hours. In that time the thief runs 22 × 0.039 = 0.858 km = 858 m.

Q.23 · Answer (D) 76
∠ABD ≠ ∠ADB, so AB ≠ AD: the equal sides must be AB = CB and AD = CD, making BD the axis of symmetry. Triangles ABD and CBD are then congruent, so ∠DCB = ∠DAB = 180° − 54° − 50° = 76°.

Q.24 · Answer (A) 5,00,000

With y votes for Y, X got 1.3y and Z 1.25y, so 0.05y = 5,000 and y = 1,00,000. Votes cast = 3.55y = 3,55,000, which is 71% of the list: 3,55,000/0.71 = 5,00,000.

English Comprehension

Q.1 · Answer (B) Depreciate

To 'appreciate' is to rise in value or to value highly; to 'depreciate' is to fall in value or to belittle.

Q.2 · Answer (A) occurrence

The correct spelling is 'occurrence', with double c and double r.

Q.3 · Answer (C) New teachers will be hired by the school.

'Will hire' becomes 'will be hired', with the object 'new teachers' as the subject.

Q.4 · Answer (C) Splendid

'Magnificent' means impressively beautiful or grand: 'splendid'.

Q.5 · Answer (B) capitol

The building where the US Congress meets is the 'Capitol'; a 'capital' is the city that is the seat of government.

Q.6 · Answer (B) able to drink

The sentence is about permission, not ability: 'you are not allowed to drink until you are 21'.

Q.7 · Answer (C) Vicious cycle

In a 'vicious cycle' (or circle) each problem makes the other worse.

Q.8 · Answer (D) well

A verb is modified by an adverb: 'She sings well', not 'good'.

Q.9 · Answer (B) POQR

'Society benefits from diversity (P) by fostering understanding and empathy (O), enriching cultures and (Q) perspectives from around the world (R).'

Q.10 · Answer (A) Apathy

Apathy is a lack of interest or concern; antipathy is dislike.

Q.11 · Answer (C) Archaeology

Archaeology studies the human past through remains such as buildings, graves and artefacts.

Q.12 · Answer (A) than the students began to leave.

'No sooner' is followed by 'than': 'No sooner did he finish his lecture than the students began to leave.'

Q.13 · Answer (A) Chaos

Chaos is complete disorder and confusion.

Q.14 · Answer (C) Beggining

The correct spelling is 'beginning'.

Q.15 · Answer (C) Incredible

'Miraculous' means so remarkable as to seem a miracle: 'incredible'.

Q.16 · Answer (C) bite the bullet

To 'bite the bullet' is to force oneself to do something unpleasant that cannot be avoided.

Q.17 · Answer (B) Accuse

To 'absolve' is to declare someone free of blame; its opposite is to 'accuse'.

Q.18 · Answer (C) A reply could have been sent by Meenakshi.

'Could have sent' becomes 'could have been sent', with the past participle 'sent'.

Q.19 · Answer (B) Implore

To 'beseech' is to beg earnestly: 'implore'.

Q.20 · Answer (D) Sovereign
A sovereign holds supreme power or authority.

Q.21 · Answer (B) simulated
AR and VR either add to a real-world environment or replace it with a 'simulated' one.

Q.22 · Answer (B) with
A virtual environment 'coexists with' the real world.

Q.23 · Answer (A) equipment
A piece of malfunctioning 'equipment', which the AR app helps to troubleshoot.

Q.24 · Answer (A) comprehensive
VR is a 'comprehensive' simulation: it replaces the user's surroundings completely.

Q.25 · Answer (A) rendition
A cartoon 'rendition' of Mike Tyson: a depiction of him.

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